

Carmen Lucca DiMario

(484) 467-2466

github.com/luccadimario

luccadimario@gmail.com

EDUCATION

Embry-Riddle Aeronautical University (ERAU)

Graduation:

Master of Science in Computer Science

GPA: 4.0

May 2027

Bachelor of Science in Computer Science

GPA: 3.97

May 2026

TECHNICAL SKILLS

Clearance: Held interim DoD Secret (inactive) | U.S. Citizen | FAA Part 107 Remote Pilot

Programming Languages: C, C++, Python, Go, GNU ASM, Bash, TypeScript

Robotics & Embedded: ROS2, PX4, Micro XRCE-DDS, NVIDIA Jetson Orin, STM32, Linux, Docker, Git

Machine Learning: PyTorch, RF-DETR, TensorRT, Stable-Baselines3, Gymnasium (PPO/SAC), NumPy

PROFESSIONAL EXPERIENCE

Garmin: Embedded Software Development Intern

May 2026 - August 2026

- Implemented Clang compiler support replacing legacy ARMcc toolchains in embedded avionics software
- Extended embedded library support to 64-bit architectures, resolving data type alignment and pointer-width issues to broaden platform compatibility
- Developed a system test suite for a proprietary communication interface on a Garmin Integrated Avionics Unit (GIA), validating real-time data transmission and protocol correctness

JRC Integrated Systems: RISE Intern/CO-OP Participant

June 2025 - May 2026

- Developed and optimized software tools for radiation-hardened semiconductor testing
- Upgraded the Accuro application's PETSc integration across a major version boundary to enable GPU-accelerated solvers for its core computations
- Rebuilt the Accuro test suite from hard-coded shell scripts into a Python framework with parallel execution, selectable test subsets, and configurable output, cutting a multi-hour run to roughly 25 minutes

PROJECT EXPERIENCE

AIRHOUND: Autonomous UAV Target Tracking [Middleware Lead]

September 2024 - May 2026

- Designed a modular four-node ROS2 perception-to-control pipeline (detection, tracking, PINN prediction, PX4 bridge) communicating with a PX4 FMUv6X flight controller over Micro XRCE-DDS
- Diagnosed DDS serialization as the pipeline bottleneck and replaced it with direct V4L2 capture, cutting mean detection latency from 234 ms to 58.8 ms and eliminating dropouts at 1280x720
- Deployed RF-DETR with TensorRT FP16 on a Jetson Orin at 15 FPS, sustaining the 10 Hz PX4 control loop
- Validated the full software stack in a 600-second SITL test at 30 Hz throughput across every pipeline stage

RoboRacer (F1Tenth) Autonomous Racing: Software Lead

August 2025 - May 2026

- Assembled and tuned the RoboRacer platform, integrating compute, sensors, and motor controller
- Brought up and configured the full ROS2 stack on the onboard computer
- Deployed and validated an end-to-end autonomy stack from sensor input through planning to actuation under real competition conditions

SEMBAS-RL: Graduate Research, Autonomous Driving Policy Assurance

June 2026 - Present

- Trained and evaluated thirteen RL driving policies (PPO and SAC) across five curriculum strategies over 5M+ training steps
- Extended SEMBAS, a three-phase adaptive boundary scanner, to RL policy evaluation over a 3D scenario space to characterize where a learned policy remains reliable
- Built three analysis agents that convert boundary geometry into ranked retraining proposals automatically

PUBLICATIONS

DiMario, C., Bacha, R., & Butka, B. "Combating Senior Scams Using a Large Language Model-Created Rubric." ASSE '24, ACM, 2024. <https://doi.org/10.1145/3702138.3702140>

Malarchick, R., DiMario, C., et al. "Predictive Target Pursuit for Autonomous UAVs Using RF-DETR with Depth-Aware State Estimation and Physics-Informed Trajectory Prediction." Proc. SPIE 14030, 140300O (2026).

<https://doi.org/10.1117/12.3094732>

EXTRACURRICULAR ACTIVITIES

- **ERAU Cyber Rodeo 2026:** First place individual
- **ERAU Division II Cross-Country/Track:** Approximately 30 hours of training weekly

<https://dimario.dev>